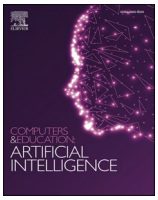
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Opportunities of artificial intelligence for supporting complex problem-solving: Findings from a scoping review

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A B S T R A C T

The research objective of this paper is to advance knowledge about the role of artificial intelligence (AI) in complex problem-solving. A problem is complex due to the large number of highly inter-connected variables affecting the problem state. Complex problem-solving situations often change decremental or worsen, forcing a problem solver to act immediately, under considerable time pressure. While research findings support the assumption that (1) affective, (2) (meta-)cognitive, and (3) social processes support complex problem-solving, opportunities of AI for supporting complex problem-solving need to be further investigated. This article presents a scoping review of relevant literature from the last five years. The study included, $N = 38$ studies for coding and analysis. Our findings show that in addition to the increased number of publications, the current trend suggests increased quality of published work. Human-AI collaboration in complex problem-solving has been explored across a broad variety of AI application domains. However, the four dimensions of complex problem – namely, cognitive, metacognitive, social and affective - have been augmented by AI to a different extent. Although most of the work has been done in the cognitive domain, it is encouraging to see progress across social and affective dimensions as well. Implications for future research and practice are being discussed.

1. Introduction

Complex problem-solving involves engaging with problems that are dynamic, could have multiple solutions, and usually require multidisciplinary teamwork (Dörner, 1980). These are "problems that resemble real-life situations" (Kunze et al., 2018, p. 3). As society has become more globalized and interconnected, the problems generated have become more complex and intractable, raising the need for skills to engage with new challenges and also a nuanced understanding of how to approach them. Fortunately, the nature of solving complex problems has changed over the last several decades, due to advances in computer science, artificial intelligence (AI), and cognitive psychology. Specifically, unprecedented amounts of data, advanced algorithms, and significantly improved computation and distributed infrastructure (such as managed data lakes) now support collaboration across diverse teams.

Working collaboratively and using technology such as an active agent to make sense of an abundance of information is a critical skill for successful learning, work, and life in general. Today's learners and workers also require the ability to navigate and astutely employ a range of social, affective, and cognitive capabilities to make sense of and address complex, real-world problems (Joksimović et al., 2020, 2022; Microsoft, 2020). In order to successfully solve complex problems, the individuals or teams involved must be able to view and simulate the dynamic problem system in its entirety, imagining the events that would take place if a particular action were to be performed (de Laat et al., 2020).

Solving complex problems, therefore, depends on the ability of individuals to integrate diverse background knowledge and expertise into a dynamic collaborative process (Van Den Bossche et al., 2011). Research on collaborative and cooperative learning and work demonstrates the importance of social and cognitive processes in creating a common understanding and shared conception of the problem and possible solutions (Van Den Bossche et al., 2011). Group work and complex problem-solving are not detached from emotions and depend on each individual's motivation and abilities to plan and monitor their activities (Hermes & Stelling, 2016). The research in group and collaborative problem-solving details the importance of understanding the interplay of factors that influence how individuals and groups regulate affective and motivational processes (Hermes & Stelling, 2016).

Recent developments in AI have the potential to support the

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integration of human and artificial intelligence and impact human and machine collaboration during group and teamwork (de Laat et al., 2020; Seeber et al., 2020; Yang et al., 2021; Siemens et al., 2022). The utilization of AI in the classroom, workplace, and life in general, has shifted from being narrowly focused on automation-based labour cost savings to augmentation of human capabilities, as exemplified by generative AI tools such as ChatGPT,[1] increasing the prospect that AI can play an active role in solving complex problems.

AI systems are capable of analysing large datasets, including unstructured data, in real-time, and detecting patterns or structures that can be used to support human decision-making and team-based problem-solving. In the context of education, AI has the potential to transform the way that learners and educators approach complex problems, by providing personalized insights and data-driven recommendations. When AI is used in complex problem solving, the patterns of interaction become more sophisticated, since AI teammates augment human contributions and can influence the resulting actions by humans. This influence of human-AI teaming increases the importance of social, affective, cognitive, and metacognitive aspects to problem-solving. Unfortunately, a substantial gap exists in the literature regarding how these attributes are expressed in joint human-AI teams in complex settings. This paper presents the findings of a scoping literature review focusing on empirical evidence on how artificial intelligence supports human complex problem-solving and the nature of human-AI collaboration in complex problem-solving at the level of (meta-)cognitive and social practices, as well as affective processes.

2. Research on problem-solving

Research on problem-solving has a long tradition in both psychology and education. Jonassen (1997) classifies problems on a continuum from well-structured to ill-structured. Well-structured problems have a well-defined initial state, a known goal state or solution, and a constrained set of known procedures for solving a class of problems. In contrast, the solutions to ill-structured problems are neither predictable nor convergent because they often possess unknown aspects. Additionally, they possess multiple solutions or solution strategies or often no solutions at all (Funke, 2012). Jonassen (2010) reiterates that the structure of a problem often overlaps with complexity: Ill-structured problems tend to be more complex, especially those emerging from everyday practice, whereas most well-structured problems tend to be less complex. The complexity of a problem is determined by the number of functions or variables involved; the degree of connectivity among these variables; the type of functional relationships between these properties; and the stability of the properties of the problem over time (Mainert et al., 2019). Simple problems are composed of a few variables, while ill-structured (complex) problems may include many variables that may interact in unpredictable ways. When the conditions of a problem change, a person must continuously adapt his or her understanding of the problem while searching for new solutions, because the old solutions may no longer be viable. Static problems are those in which the factors are stable over time while ill-structured problems tend to be more dynamic (Seel et al., 2009).

Accordingly, a problem is complex due to the large number of highly inter-connected variables affecting the problem state (Eseryel et al., 2013). This means that changes in one variable influence the status of many other related variables; therefore, it is difficult to anticipate all possible consequences of any action. Not all of the variables lend themselves to direct observation. Dealing with the intransparency of the problem variables and the interrelationships among them is often difficult due to time-delayed effects, i.e., not every action shows immediate consequences. Complex problem-solving situations often change decrementally or worsen, forcing a problem solver to act immediately, under considerable time pressure, such as determining the causes of overpopulation, hunger, and pollution. Therefore, complex problem-solving situations bear multiple goals, some of which could be contradictory requiring a reasonable trade-off. A complex problem is ill-structured because such problems "cannot be described with a high degree of completeness or solved with a high degree of certainty" (King et al., 1994, p. 10). All of these factors make ill-structured problem--solving very challenging (Funke, 1991).

One of the most developed areas of research in AI is centred around supporting various processes involved in complex problem-solving. As such, contemporary research spans various domains, from healthcare and augmenting medical decision-making, supply chain optimization, to environmental monitoring analysing various sources of data, including various sensors and satellite images to disaster response. One common characteristic, despite the domain of complex problems, is that AI is primarily used to support human capabilities. Current research and practice in AI mainly focuses cognitive support by processing large amounts of data and summarizing information from various sources, helping humans to identify emerging patterns, and make decisions. However, complex problem-solving is a multidimensional construct that also involves affective and social processes (Döppner et al., 2019; Eseryel et al., 2014; Hermes & Stelling, 2016). This study is therefor aimed at reviewing literature to understand how AI support human complex problem-solving at the level of affective, (meta-)cognitive and social processes.

3. Opportunities of artificial intelligence for complex problem-solving

Building on the previous work by Andrews-Todd and Forsyth (2020), Salas et al. (2008), Barth and Funke (2010), Dörner and Funke (2017), and Spering et al. (2005), among others, we posit that complex problem solving encompasses three interrelated domains of (***meta-)cognitive***, ***affective***, and ***social*** processes. Although there is an abundance of research around supporting **cognitive** processes of complex problem-solving using AI, such as decision-making, there is a lack of evidence linking AI and other domains of complex problem-solving, namely social and affective (Osbeck & Nersessian, 2011). Specifically, the existing research on augmenting human capabilities using AI predominantly discusses the division of cognitive tasks between humans and machines, using machines in recognition of human affective states, or exploring how machines support social activities, such as co-creation (Fu & Zhou, 2020). Such division of roles between humans and machines is rather static and is usually performed on arbitrary assumptions about the expected performance and the notion of who does what better in the context of mixed human-AI partnership (Seeber et al., 2020). In the context of supporting cognitive processes of complex problem-solving, it is crucial to observe this domain as an emerging, rather than static (de Laat et al., 2020). That is, the cognitive dimension of human engagement unfolds in various ways across the phases of complex problem-solving (e.g., brainstorming, data collection, decision-making). Therefore, it is critical to establish roles, actions, and capabilities needed for an AI partner to coordinate complex problem-solving practices. This is, however, equally important for metacognitive, social and affective domains that seem underdeveloped in contemporary research.

A growing body of empirical studies shows that complex problem-solving is highly related to ***affective*** states (Pekrun et al., 2014; Spering et al., 2005). In Berlyne's (1971, p. 336) terms, the novelty and complexity of a problem-solving situation can induce a cognitive dissonance that in turns increases arousal mediating an affective reaction, for instance, curiosity or anxiety. Complex, novel, or incongruous objects have in common that they resist immediate assimilation and evoke emotions that interact reciprocally with cognitive appraisals (Kuhl, 2000), and thus, create a basis for goal-directed actions (Gross, 1998). Findings of experimental research show consistently that positive

[1] https://chat.openai.com/chat.

state emotions are more likely associated with the use of heuristic processing strategies that are characterized by the productive use of schemas, discussed in terms of generic knowledge structures, and related top-down processing (Ifenthaler, 2015; Spering et al., 2005). In contrast, negative state emotions are more likely associated with bottom-up processing and a more systematic gathering of information as well as with paying more attention to the details of the problems to be mastered (Fiedler, 2001). This further suggests considerable opportunities for the use of AI in the context of emotional-driven adaptive systems for decision-making support or using NLP to identify patterns and trends in the relationship between emotions and language use (Tillman & Louwerse, 2018).

A general assumption in problem-solving research is that the first thing a person does when confronted with a problem is to try to construct a mental representation of its relevant features (Seel et al., 2009). This internal representation of a problem is termed a *problem space* – and if its construction has been successful the problem space will consist of information about the initial and goal state of the problem as well as information about the operators which can be applied to solve it. More specifically, Rumelhart et al. (1986) argue that these internal functions during problem-solving are dependent on two interacting modules or sets of units: (1) schemata and (2) mental models. The resulting cognitive architecture corresponds to a great extent with Piaget's epistemology (Piaget, 1976) and its basic mechanisms of assimilation and accommodation (Ifenthaler & Seel, 2011). Assimilation is dependent on the availability and activation of schemata, which allow new information to be integrated immediately into pre-existing cognitive structures. If a schema does not fit immediately with the requirements of a new task it can be adjusted to meet them using accretion, tuning, or reorganization (Seel et al., 2009). Accordingly, if a schema for any problem type is available, the schema is mapped onto the problem to be solved promptly (Jonassen, 2000). If assimilation is not successful, accommodation must take place to reorganise or restructure an individual's knowledge. However, when no schema is available or its reorganization fails, the human mind switches to the construction of a mental model which is defined as a dynamic ad hoc representation of a phenomenon or problem that aims at creating subjective plausibility through simplifying and envisioning the situation, or through analogical reasoning. From a ***meta-cognitive*** perspective, the above-mentioned processes may be monitored and reflected for advancing these functions for future events (Azevedo et al., 2013).

Finally, shared mental models are regarded as a significant factor for successful problem-solving (Cannon-Bowers & Salas, 2001). However, the concept of shared mental model is used and interpreted differently by various disciplines, for example, industrial and organisational psychology, human factors, social psychology, or system dynamics (Badke-Schaub et al., 2011). Previous research on **social** processing in complex problem-solving shows that if humans share similar individual mental models they are more effective in their problem-solving performance (Van Den Bossche et al., 2011). For example, Lim and Klein (2006) found that shared task knowledge and shared team knowledge were valid predictors for problem-solving performance. Similar findings regarding the influence of shared task and team knowledge on problem-solving performance were found in a series of studies using flight simulators in laboratory settings (Mathieu et al., 2005).

While research findings support the assumption that (1) affective, (2) (meta-)cognitive, and (3) social processes support complex problem-solving, opportunities of AI for supporting complex problem-solving need to be further investigated [18]. Thus, the central research objective of this paper is to advance knowledge about the role of AI in complex problem-solving. To guide the scoping review, the following research question was formulated: How may AI complement humans during complex problem-solving through the functions of (1) affective processing, (2) cognitive and meta-cognitive processing, as well as (3) social processing?

4. Method

4.1. Data collection

Given the rather heterogenous nature (i.e., draws on various fields and disciplined) of the research on human-machine collaboration and complex problem-solving, we opted for a scoping review, instead of a more precise systematic literature review. In so doing, we still followed the guidelines for a scientifically rigorous literature review, except that we searched only one digital database – i.e., Scopus, following an approach similar to Cetindamar et al. (2022). In general, scoping reviews are commonly used for "reconnaissance" - to clarify working definitions and conceptual boundaries of a topic or field. Scoping reviews are therefore particularly useful when a body of literature has not yet been comprehensively reviewed or exhibits a complex or heterogeneous nature not amenable to a more precise systematic review of the evidence. Moreover, compared to other databases, such as Web of Science, Scopus also includes conference papers that are crucial for emerging fields such as human-AI collaboration. Compared to Google Scholar, on the other hand, Scopus includes less grey literature, and it has been widely used in bibliometric studies (Caputo & Kargina, 2022; Tibaná-Herrera et al., 2018).

Given our focus on understanding the dimensions of human-AI collaboration in complex problem-solving, we defined the following search query in Scopus:

TITLE-ABS-KEY (("problem solv*") AND ("artificial intelligence" OR "machine learning"))

We intentionally opted for "problem-solving" as a basis for our search, rather than "complex problem-solving", to capture a broader set of studies, as there might be a difference in conceptualisations between ill-defined and well-defined problems across diverse fields of research. We limited our search to include studies published over a five year period, starting in 2017 resulting in 2607 studies. After removing duplicates, we obtained 2597 studies considered for screening. The screening process was conducted by two researchers who read the title and the abstract for each of the studies, resulting in 809 studies considered for inclusion. Both researchers have been actively exploring the domain of human-AI collaboration with previous work being done in developing models for complex skills assessment, such as creativity, critital chinking or problem solving. Given the complexity of the topic, two researchers skimmed all 809 studies and reviewed them for inclusion or exclusion, discussing every decision until a common agreement was met. The inclusion criteria was defined around the following principles: only studies that are relevant to the research question and published as peer-reviewed journal articles, conference proceedings, books, or reports will be included; only studies published in English will be included in this scoping review; studies published from 2017 to 2021 were considered for inclusion; studies must include human participants, regardless of age, gender, ethnicity, or geographical location. The level of agreement between coders was generally high, with an accuracy rate exceeding 95%. However, there were some instances where the coders disagreed, particularly in cases where the studies employed different tools or simulations to facilitate teamwork. All disagreements were resolved through discussion and consensus among the coders. The majority of the studies that were excluded were either focused on developing an AI algorithm, rather than human-machine teaming, were not focusing on supporting humans in complex problem-solving, or were not empirical studies (e.g., an opinion paper discussing the nature of human-AI collaboration in complex problem solving). The final dataset included **38 studies** selected for coding and analysis. The list of selected papers has been explicitly specifically marked in the reference list.

4.2. Analysis

To address our research questions, two coders, also authors of this paper, coded all 38 studies based on the following dimensions.

- General information: publication year, document type (e.g., conference or journal paper), country.
- Author information: author subject area (e.g., computer science, phycology, education), author research interests (e.g., computer vision, human-computer interaction). Information about the author's subject area was collected from Scopus whereas the research interests were obtained from authors' public pages (e.g., their institutional profile page).
- Study design details: sample size, context (e.g., higher education, workplace), data sources (e.g., multimodal, click stream), research questions, the context for problem-solving (e.g., healthcare, military teams), research type (i.e., research, conceptualisation, evaluation, or implementation), method (e.g., experimental, exploratory analysis), limitations reported (yes or no).
- Domains of human-AI collaboration: cognitive (e.g., cognition, exploration), social (e.g., collaboration), affective (e.g., trust, confidence), and metacognitive. The four domains were determined by analyzing the existing literature on complex problem solving, as described in Section 3. Two coders used inductive coding to identify concepts mentioned in the studies included in the review and categorized them into the four domains – cognitive, social, affective and metacognitive. After both coders identified relevant terms, all the terms were combined and discussed with the larger group of authors. This approach was chosen because the field of human-AI partnership is constantly evolving, and there is no established taxonomy.

We used this data to provide a quantitative overview of the current state of research, discussing the number of publications across different research fields or industries, the count of publications per year, including the sample size overview, as well as the nature of the studies, following the same categorisation as the one introduced in Dawson et al. (2014) (i.e., research or theory development). Finally, we provided a qualitative summary of the four main domains of human-AI collaboration, namely cognitive, metacognitive, affective and social.

5. Results and discussion

5.1. Summary of key publications

Although the number of publications in the area of human-AI collaboration dropped in 2018 and 2019, it seems that over the last two years, this topic continued to receive considerable attention (Fig. 1, left side). In addition to the increased number of publications, the current trend suggests increased quality of published work. Specifically, Fig. 1 (left side) shows a sustained increase in the number of journal publications since 2018. Moreover, it is also noteworthy that recent publications tend to utilize a considerably larger sample size (Fig. 1, right side). While early publications in this domain either did not report sample size (or it was not relevant – e.g., conceptual papers), over the last two years, it seems that recent contributions tend to be research focused with large numbers of participants.

This is also well supported by the trend presented in Fig. 2 which shows an increase in the number of empirical research published on the topic. Given the novelty of the field, a higher ratio of conceptual papers is expected. This type of publication invites broader discussion on the topic of human-AI collaboration and introduces various perspectives. Empirical research is needed to further evaluate those assumptions in practice. The lack of theory development work, however, is concerning. To advance the field of human-AI teaming, it is necessary to develop a theory that will provide a common language, shape the key objects of study and units of analysis, and highlight relevant issues (Damşa & Jornet, 2021; Kaptelinin & Nardi, 2006). Nevertheless, over the last five years, only two attempts were recognised that aimed at contributing to developing a theory of human-AI collaboration in the context of complex problem-solving (Fig. 2).

Human-AI collaboration has been explored across a broad variety of AI application domains. Healthcare represents arguably the most common domain of study of human-machine collaboration (Fig. 3). In this context, human-AI collaboration is primarily considered within clinical decision-making, such as rehabilitation assessment or skin cancer recognition, and healthcare assistance, such as using social robots for in-patient care (e.g., Cobianchi et al., 2023; Tschandl et al., 2020). The use of social robots is not unique to healthcare sector, as shown in Fig. 3. Specifically, social robots are also being used in educational context, as teaching assistants or voice assistance in K-12 settings (Guggemos et al., 2022). Another large domain of exploration of human-AI collaboration is in supporting design creativity across primary, secondary, and lifelong learning settings. For instance, AI has been used in co-creating interactive arts or supporting K-12 students in creative writing exercises (Sharples, 2022). Likewise, the adoption of AI in workplace and human-AI collaboration have also been explored in the context of supporting criminal investigations or supporting decision making in air cargo industry. Finally, the advanced human-machine symbiosis and successful behaviours of hybrid human-AI teams have been dominant areas of research in the military domain.[2]

The exploration of human-machine collaboration is interdisciplinary by nature. As further shown in Fig. 3, the wide variety of application domains necessities multidisciplinary and interdisciplinary efforts to understand and address the challenges of successful human-machine collaboration. Therefore, it is important to highlight that authors currently publishing in the domain come with subject area expertise that spans across a variety of disciplines, ranging from social and environmental sciences to computer science and mathematics (Fig. 4). Specifically, besides a considerable number of authors with computer science (each publication in this review included at least one author with computer science expertise) and engineering background, 17 authors have expertise in social sciences (Fig. 4). Finally, given the emergence of AI in healthcare, it seems expected that more than 20 authors are coming from medicine, neuroscience, or psychology background (Fig. 3).

Although research interests in AI, Human-Computer Interaction (HCI), and machine learning represent the dominant focus of authors of included publications, only nine studies included collaboration between authors who explicitly labelled their research interest in at least two out of those three research areas (Fig. 5). On the other hand, 18 publications had at least one author with research interests in AI, HCI or machine learning, whereas 11 publications had no authors with expertise in these three areas (Fig. 5). Other dominant research domains include computer vision, engineering design and human-robot interaction, whereas several authors explicitly stated their research interests in neuroscience, robotics and social robotics (Fig. 5). Finally, at least one publication included authors with research interests in research domains such as education, explanation-aware computing, innovation and arts, legal psychology, ethics, management, sport, tourism or immunology, to name a few.

5.2. Dimensions of human-AI collaboration

Despite the obvious diversity in disciplinary domains addressing the problem of human-AI collaboration in complex problem-solving, much of the work in examining human-machine partnership, and the dominant conceptions surrounding AI, remains rooted in the processes and developments of the automation of routine tasks or the augmentation of the cognitive or physical capabilities of individuals (Ekbia & Nardi, 2017; López-González, 2020). Most studies included in this review focus on understanding and explaining a cognitive dimension of human-AI partnership. The affective domain seems to be well represented in contemporary literature, as well. This line of research builds on the emergence of affective computing (Caruelle et al., 2022) and Emotion AI (Monteith et al., 2022). Social processing, on the other hand, remains at

[2] https://airpower.airforce.gov.au/videos/human-machine-teaming.

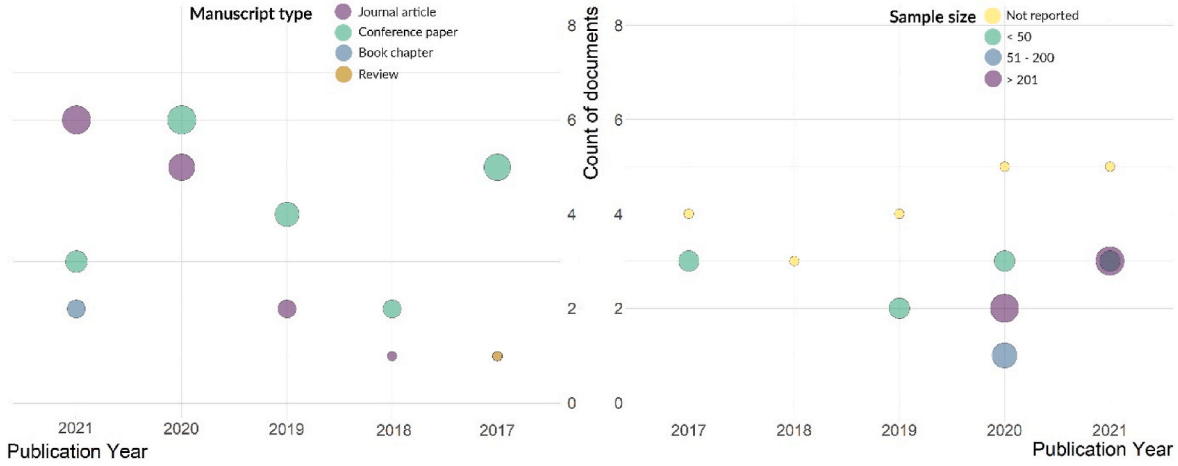


Fig. 1. Distribution of documents per year, where the left side of the figure shows the count per document type (i.e., journal article, conference paper, book chapter, or review), and right side shows the count per sample size.

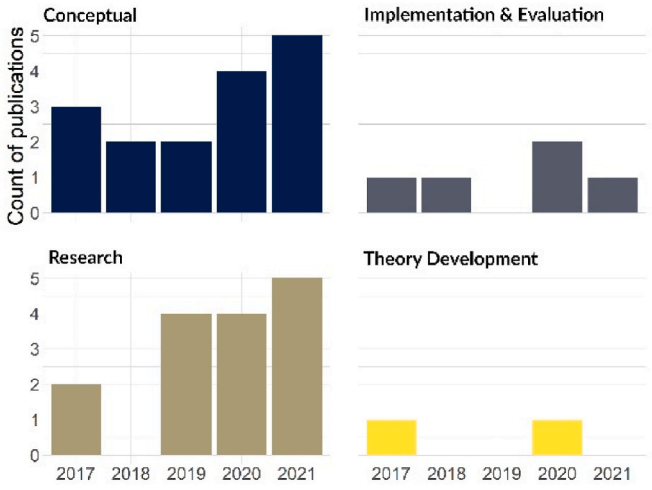


Fig. 2. Count of publications for each of the publication types (i.e., conceptual, implementation & evaluation, research, and theory development) across years.

a very shallow level of exploration, whereas the metacognitive dimension has not been explored in the current literature.

5.2.1. *Cognitive*

Coding 38 studies included in this review, we identified the following cognitive tasks/dimensions as being commonly supported (or fully automated) by AI.

- **Recognition**, such as pattern recognition or search and profile.
- **Action**, such as knowledge consumption or talk.
- **Cognition**, including knowledge representation, shared mental models, and statistical reasoning.
- **Exploration**, such as knowledge structuring.
- **Control**, including control over task performance or supporting humans' goals and intentions.
- **Decision-making**, across a variety of domains.
- **Recommendation/Forecast**, and
- **Creativity.**

The cognitive tasks outlined above (e.g., recognition or action) have not been observed in isolation. In fact, the majority of studies observed a combination of cognitive activities, mostly in the context of supporting human decision-making. For instance, in an attempt to differentiate activities that machines do well from those that are implicitly human, Kozlova et al. (2021) posit that, in the ***decision-making*** process, ***forecasting*** should be offloaded to machines whereas ***judgment*** needs to remain exclusively human activity. Kozlova et al. (2021) continue with a claim that judgments "require an understanding of the outcome of any interaction" (Kozlova et al., 2021, p. 615). However, humans are not able to match machines in activities such as speech recognition, aggregation of information from sensors, computer vision, gesture, biometric, and character recognition (Kozlova et al., 2021, Hayashi & Wakabayashi, 2017, February).

Introducing the notion of hybrid-augmented intelligence, Zheng et al. (2017) define human intelligence as creative, complex and dynamic. In that sense, human intelligence is unmatched in abstract thinking (*cognition* dimension, as outlined above), reasoning (*decision--making*), innovation and creating new knowledge (***action***), as well as making associations (***recognition***). Machines, and AI specifically, have the features of normalization, repeatability, and logicality (*recognition* and *action*). Likewise, Cesta et al. (2018) presented an AI-based cognitive architecture for assisting older persons in their daily routines. The proposed architecture utilised AI in integrating sensor data, knowledge representation (***cognition***), and automated planning techniques (*recognition*).

Although contemporary research argues that ***creativity*** is a fundamental dimension of human intelligence (Boden, 1998; Zheng et al., 2017), this cognitive dimension has attracted considerable attention in human-machine collaboration research. From augmenting design creativity (Chan & Lee, 2020; McCormack et al., 2020; Zeng et al., 2019; Zhang et al., 2021; Zhou et al., 2020), creating interactive arts with AI support (Chen et al., 2020), to the development of the art therapy robots (Cooney & Menezes, 2018) and utilizing robots in co-creative writing exercises (Kantosalo & Riihiaho, 2019), AI has been perceived as a valuable team member in fostering human creativity. As such, current research covers all three approaches in using AI in creating new ideas, as posited by Boden (1998): "by producing novel combinations of familiar ideas; by exploring the potential of conceptual spaces; and by making transformations that enable the generation of previously impossible ideas" (Boden, 1998, p. 347).

5.2.2. *Affective*

Within the affective domain, the following dimensions have been considered as being crucial for successful human-AI collaboration.

- **Emotion**, including variables such as social anxiety, fun, and emotional states in general,

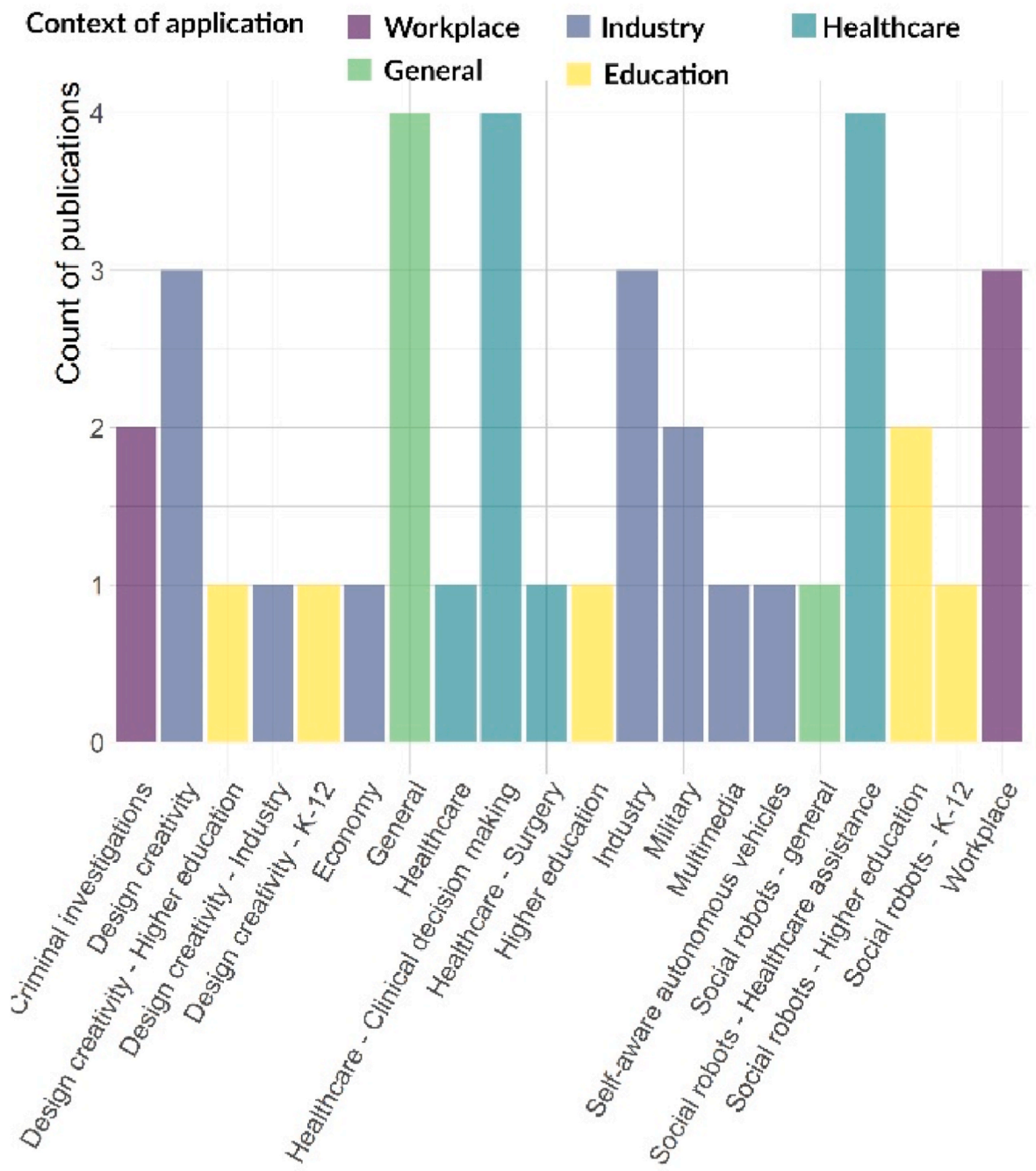


Fig. 3. Domains of AI applications that served as use case scenarios for the exploration of human-AI collaboration.

- **Attitudes**, mainly focusing on sense of relaxation, usability, or outcome satisfaction, among other dimensions,
- **Trust**, including transparency and interpretability, as well as dimensions such as distrust and mistrust,
- **Confidence**, such as confidence in AI or performance expectancy,
- **Saliency**, and
- **Moral.**

The role of human ***emotions*** in developing a partnership with AI has been observed from several aspects. For instance, Zhu and Deng (2021) explore the association between social anxiety and the adoption of a robotic training partner, showing a tendency for students with higher social anxiety to choose a robotic trainer partner over human partners. Designing an art therapy robot, Cooney and Menezes (2018) developed a robot capable of expressing emotions through colour, composition and textures. Despite an ongoing debate about whether robots are capable of sensing different emotional states (e.g., Chuah & Yu, 2021), this line of research showed promising results in enabling machines to "convey" complex emotions. The majority of work in this area, however, focuses on developing AI-enhanced systems for sensing human emotional states in order to provide the necessary support.

To understand humans' ***attitudes*** that shape their interactions with AI, authors explored a variety of variables, mostly framed around general usability and outcome satisfaction. For instance, Zhu and Deng (2021) explored to what extent a human's intention to adopt a robotic training partner is associated with a sense of relaxation. Lee et al. (2021), on the other hand, collected data about therapists' attitudes towards the usefulness, intention to use, workload (e.g., how difficult it

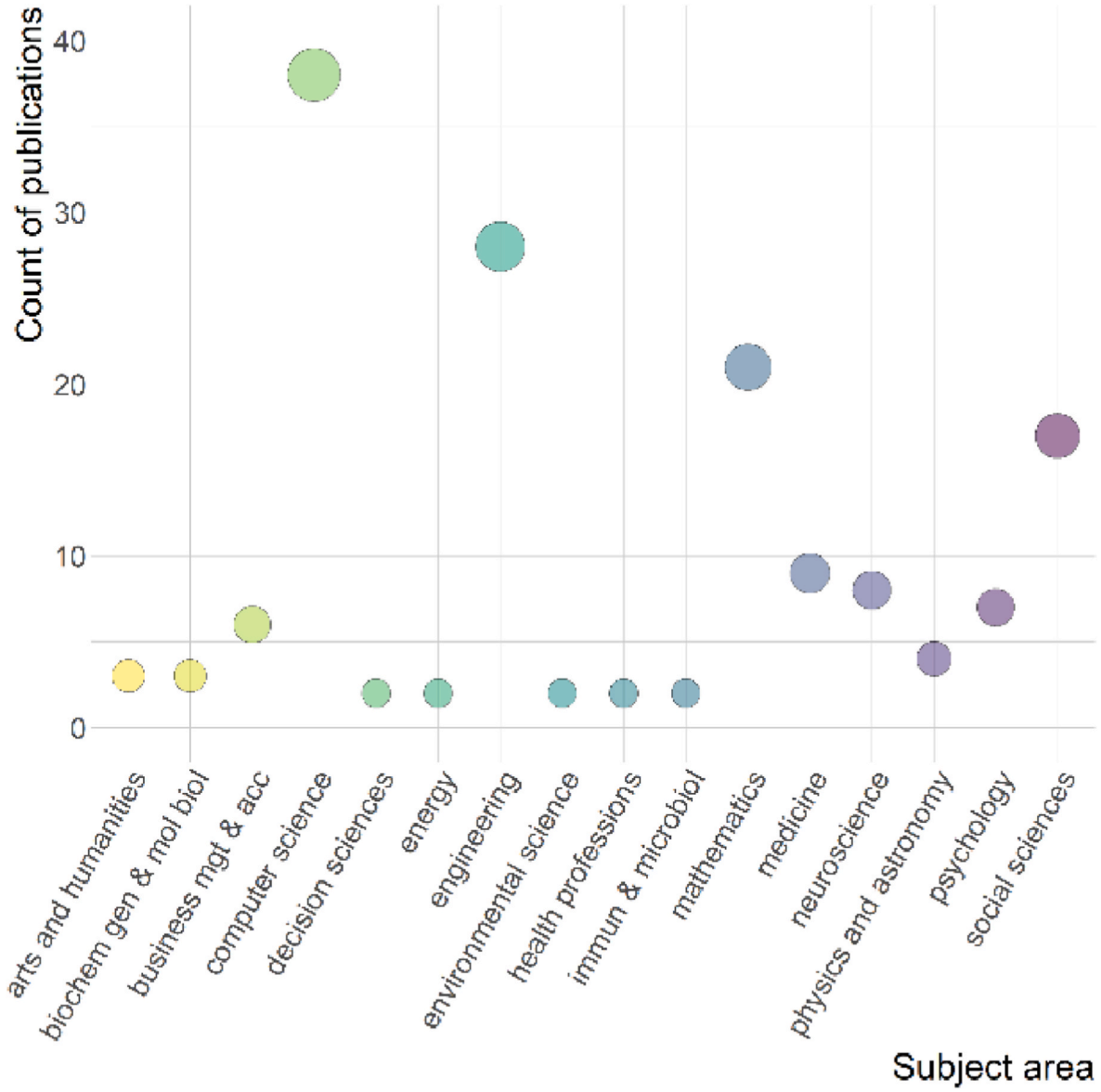


Fig. 4. Count of publications including authors from a given subject area.

was to accomplish a task in collaboration with an AI partner) and richness (in functional capabilities) when collaborating with an AI partner in clinical decision making (Moreno et al., 2017; Nothdurft et al., 2017; Van Baalen et al., 2021). Outcome satisfaction, ownership and expressiveness were explored in the context of co-creative writing exercises (Kantosalo & Riihiaho, 2019). Finally, aesthetic feeling and the notion of increased fun were explored across diverse workplace settings, such as design and video gaming (Chan & Lee, 2020; Pfau et al., 2020).

The notion of ***trust*** has been identified as one of the driving mechanisms of successful human-machine collaboration. Similar to human collaboration, effective partnership with machines also requires trust and accountability (Grigsby, 2018; Hepenstal et al., 2021; McCormack et al., 2020). Dubey et al. (2020) differentiate *calibrated* trust from *interpretability*. Calibrated trust is defined as the extent to which humans trust machines and in which particular situations. Interpretability, on the other hand, represents humans' ability to interpret the behaviour of an AI assistant (Dubey et al., 2020). Such differentiation has potentially immense implications for research and practice as it clearly points to the importance of contextualizing AI (Vercauteren et al., 2020). That is, having confidence (or trust) that AI will reliably find a route from one location to another is substantially different from the situation where humans and AI collaborate in a neurosurgical intervention. For instance, in the context of human-AI collaboration in clinical decision-making on rehabilitation assessment, Lee et al. (2021) showed that therapists tend not to trust AI systems that are not interactive (i.e. black-box solutions). Arora et al. (2021) further unpack the notion of trust and discuss generalization, explainability, interpretability, risk mitigation, fairness, and inclusiveness as key pillars in making "AI-based decisions in medical and public health settings" (*ibid.*, p.11). Whereas Van der Waa and colleagues (2021) focus on the ***morally*** salient tasks, arguing for the importance that human partners still need to exert meaningful control over the artificial teammate.

5.2.3. Social

In the studies included in this review, human-machine relationship has been commonly described as.

- **Collaboration** in decision-making or creative work, for example,
- **Cooperation**, such as co-creation in design and creative work and learning, or
- **Assistance,** such as in-patient care or daily home support.

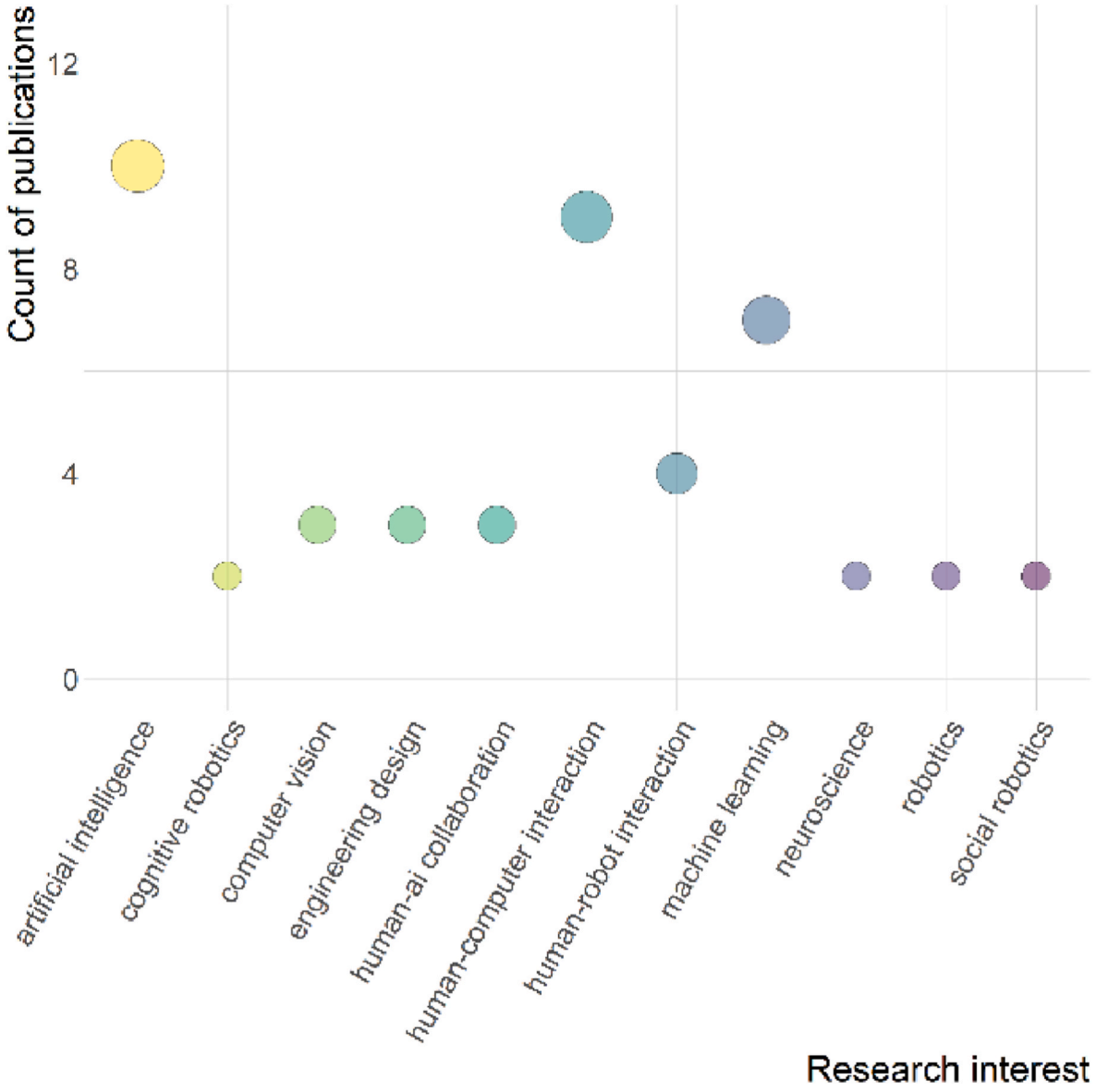


Fig. 5. Count of publications including authors with a given research interest.

Collaboration has been primarily observed in the context of supporting human decision-making using AI. Zheng et al. (2017) provide a comprehensive overview of relevant use cases that depict human-AI collaboration in managing industrial complexities and risks, collaborative driving, or healthcare, among others. Vercanteren and colleagues (2020) extend the discussion on the use of AI in healthcare settings by observing collaboration between humans and machines, as well as between machines (robots). It is noteworthy various aspects of collaborative creativity. For instance, Chen et al. (2020) define collaboration and coordination and synchronization. To produce quality interactive and collaborative artworks, humans and machines need to maintain coordination and synchronization (Chen et al., 2020). Finally, Kantosalo and Riihiaho (2019) showed that the quality of creative ideas was higher when these assumed input from both humans and machines, rather than machines only.

Nevertheless, productive collaboration between humans and machines depends heavily on the cognitive and affective capabilities of AI. As Bortnikov and Samsonovich (2018) argue, for a machine to engage in productive collaboration, two necessary preconditions need to be met. First, machines need to be able to *understand* the "developing narrative in which they participate" (Bortnikov & Samsonovich, 2018, p. 17). Therefore, it comes as no surprise that most of the recent development in AI has been evidenced in DALL-E 2,[3] GPT-3,[4] and BERT.[5] This discussion took another turn when a Google employee recently claimed that he was convinced that an AI chatbot was sentient.[6] Although AI is still relatively far from reaching singularity (Wood, 2022), it seems highly unlikely that another AI Winter is coming any time soon as well. Second, to become a collaborative partner, AI needs to react adequately to human emotions (Bortnikov & Samsonovich, 2018). As our findings suggest, this seems to be one of the most challenging domains in the further development of human-AI collaboration in complex problem-solving.

Cooperation represents one of the main principles of human communication and interaction with other humans (Canan et al., 2017; Kohn, 1992; Lemaignan et al., 2017; Lindenfors, 2017). Human

[3] https://openai.com/dall-e-2/.
[4] https://openai.com/blog/gpt-3-apps/.
[5] https://towardsdatascience.com/bert-explained-state-of-the-art-language-model-for-nlp-f8b21a9b6270.
[6] https://www.scientificamerican.com/article/google-engineer-claims-ai-chatbot-is-sentient-why-that-matters/.

cooperation is commonly perceived as a multidimensional construct comprised of both "social-cognitive skills for creating the joint intention and attention needed in verbal communication and the prosocial motivations needed in social interactions to help and share with others" (Aeschlimann et al., 2020, p. 2). Aeschlimann et al. (2020) further show that communication with artificial agents (such as voice assistants) is drastically different from communication and cooperation with humans. Specifically, children tend to share less with a voice assistant, than with their peers, regardless of whether this AI assistant has human-like features or not. Huber and Weiss (2017), however, show that such a finding does not hold in a factory (industry) environment. Opposite to the findings in the K-12 settings, industry employees were more engaged with a robot featuring physical human-robot interaction (Huber & Weiss, 2017, Lawless, 2021).

The application of AI in healthcare and other industries (such as tourism) has been primarily framed around the ***assistance*** to in-patient care, daily-home support, and clinical care (Cesta et al., 2018, pp. 3–15; Conti et al., 2020; Kriegel et al., 2019; Perić & Vitezić, 2021). For instance, Cesta et al. (2018) and Kriegel et al. (2019) explore the application of AI for the support of older people, showing that socially assistive robots have a wide range of applications from cleaning and transporting food cart to helping with food selection and assist with biography work. Conti et al. (2020) further show that children with ASD tend to prefer to interact with robots rather than non-robotic instruments or humans. Likewise, recent research also shows that humans tend to have more trust in human-like AI (Oksanen et al., 2020). Such findings have strong implications for designing AI-enhanced interfaces whether those being robots or simple chatbots.

5.3. General discussion

AI is rapidly reshaping the relationship that exists between humans and machines. The prominence of AI is growing exponentially and is increasingly influencing how we learn and how we undertake our work (de Laat et al., 2020; Siemens et al., 2022; Yang et al., 2021). Globally, investments and research funding are targeted at developing smart automated solutions, but attention to how artificial systems are integrated with human groups is largely overlooked. For example, national AI strategies around the world focus on important topics such as bias, ethics, policy, and increasing capacity through skill development. Lacking, however, is a focus on how basic human-AI interactions are coordinated when engaged in knowledge work such as solving complex problems.

In that sense, MIT Sloan recently proposed a comprehensive framework that describes five human-AI interaction modes, namely: i) automator, ii) decider, iii) recommender, iv) illuminator, and v) evaluator (Ransbotham et al., 2020). Such division perfectly reflects the current state of AI and aligns with previous work that focuses on supporting human cognitive capabilities. In this scoping review, however, we posit that as AI further evolves towards General AI, other dimensions of human-AI collaboration will require significantly more attention.

Fig. 6 outlines the main findings of our scoping review, showing the extent to which each of the four dimensions of complex problem solving has been augmented by AI. It is important to note that the four dimensions are not independent of each other. Those four domains need to be observed interchangeably, as interconnected dimensions, which is usually the case when studying human collaboration in complex problem-solving. Likewise, a comprehensive model of human-AI collaboration is necessary to provide a common language, shape the key objects of study and units of analysis, and highlight relevant issues in the domain.

Contemporary research in human collaboration, group work and teamwork domains, also shows that metacognition and affect precede the observable cognitive and social changes (Hutchins, 2006). To our

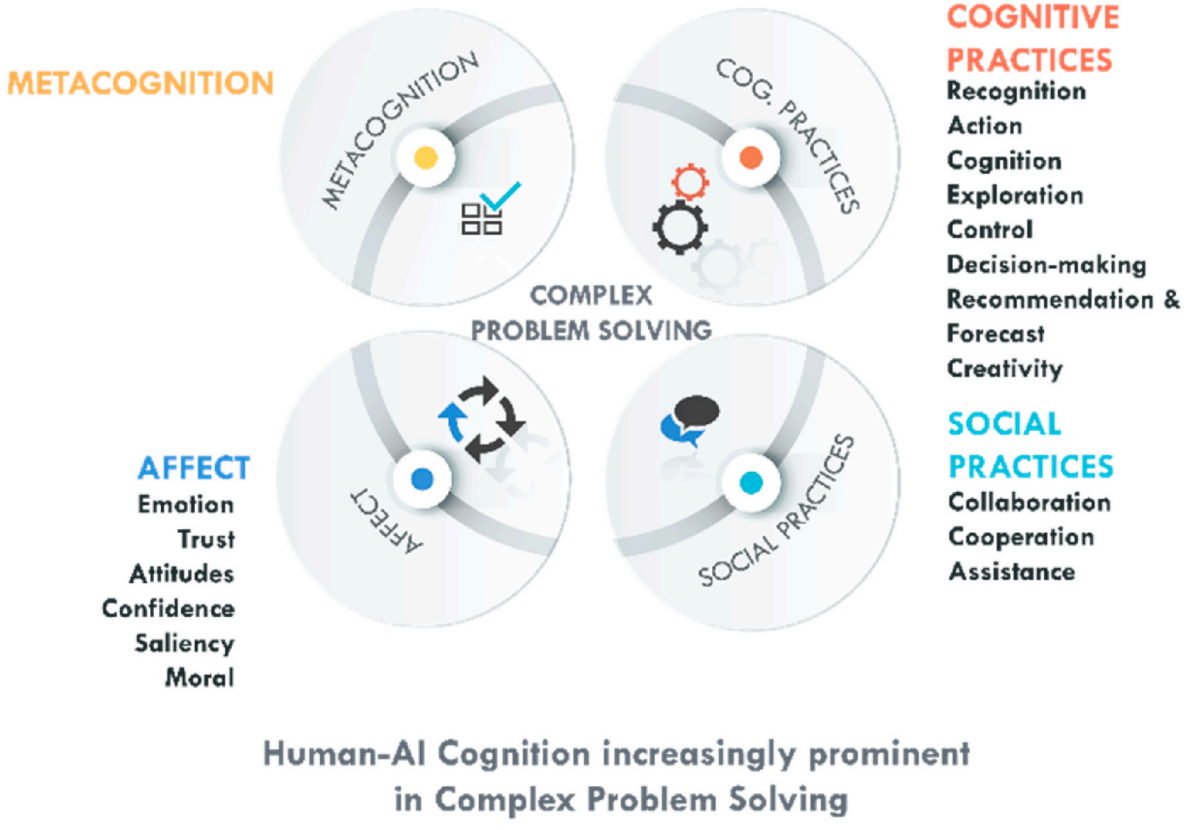


Fig. 6. Domains of Human-AI collaboration, as evidenced in the contemporary literature. Note that in our scoping review, we failed to identify studies that would explore human-AI interaction at the metacognitive level.

knowledge, no research in complex problem solving tackles the relationship between these four constructs, let alone how this impacts joint human-AI teams. Despite extensive literature around distributed cognition, teamwork, and problem solving (Canan et al., 2017; Dörner & Funke, 2017; Hutchins, 2006), no study to date attempts to theorise the interplay between human and artificial collaboration in complex problem-solving at the level of (meta-)cognitive and social practices and affect. This perhaps comes as no surprise, as AI research is yet to produce technology that is capable of fully engaging in the cognitive (e.g., a shared mental model), and social (e.g., trust and relational) processes needed for sensemaking in solving real-world problems. As argued by Klein et al. (2005), to become a "real" team player (on par with humans) AI needs to be able to: i) engage in common ground activities; ii) be reasonably predictable and able to predict actions of others; iii) engage in goal negotiation; iv) coordinate activities with humans; v) interpret and send intention signals; and vi) control the costs of coordinated activity. Although rapid progress is being made in computer science and related fields towards these goals, a key deficit militating against such advancements relates to the lack of research looking into how humans collaborate with AI in sensemaking. For instance, in their recent study Gyory et al. (2021) analysed discourse data to show that communication in AI-assisted teams was richer, more diverse and more frequent, compared to teams that did not have AI assistance. While this work represents an important contribution to the development of science of human-machine teaming, the study was conducted in a structured and specifically designed environment that allows for a limited number of team dynamics to be observed (Gyory et al., 2021). This proposal posits that theory and a rigorous conceptual framework are necessary precursors to technical development particularly in social domains, such as human learning or complex problem-solving.

When it comes to educational research and practice, a critical theme going forward will be the design of assessment, which drives teacher and student behaviour. More than 30 years ago, Salomon and his colleagues (1991) raised the issue of whether to evaluate the performance of the student/machine system as a whole or just the student after the AI is removed. The emergence of generative AI has prompted educational institutions to urgently review their assessment design, as AI writing applications are now capable of producing polished writing that can partially or fully answer writing assignments across various fields, provided with the appropriate prompts. Although designing with humans-in-the-loop is often praised as a principle, this assumes that humans are competent. However, there is currently little evidence that students are capable of engaging critically with generative AI applications. To make effective use of generative, conversational AI, this knowledge gap must be filled rapidly, building on prior research in automated writing evaluation, conversational user interfaces, and pedagogical agents.

6. Conclusions and future research

Human interaction is a complex phenomenon that is dynamic and multimodal by nature (Levinson, 2006). Team interactions are multimodal as they include facial expression, speech, body orientation and sometimes tool manipulation, depending on the context (Hutchins, 2006; Yang et al., 2021). As robots and AI become more ubiquitous in our communication and interaction, novel theories and conceptual models of human-AI collaboration are needed. As Levinson explains, studying human interaction spans a variety of disciplines, including education, anthropology, sociology, biology, and psychology. Adding machines to the situation brings another layer of complexity that calls for the utilization of methods and approaches emerging from human-computer interaction, computer-supported collaborative learning and work, statistics, data science and related domains (Blasch et al., 2019; de Laat et al., 2020; Gyory et al., 2021). Researching human-AI teaming requires (re)framing the core object of investigation, one that moves from the arbitrary division of roles based on the current understanding of who (humans or machines) performs which task better. This division of roles, while useful at the current level of AI development, does not provide a means for understanding a complex symbiosis between humans and machines as they interact in complex problem-solving. This approach, that involves fixed division of roles, neglects the emergent character of interaction and the negotiation of roles and tasks during complex problem solving. For effective augmentation of human intelligence by machines, the roles of each must emerge through interaction.

It is important to note that study of human-machine teams involves studying interactions between two dynamic systems, that spans across four domains – cognitive, metacognitive, affective, and social. That is, humans will adapt what they do as they learn to make better use of what the AI has to offer. Correspondingly, the AI will (over longer periods of time) learn to do a better job of understanding and meeting the needs of the human team members. In other words, the sheer complexity of research on human-machine teaming stems from the fact that we are observing two dynamic systems, because both parts (human and AI) will be changing. We have no reason to believe that what either of the partners in human-machine teams does – and therefore how they interact – will stabilise. Future research needs to provide a comprehensive description of relevant cognitive, social, and affective dimensions, and how these unfold throughout the problem-solving process. The outcomes of this potential work under this domain need to identify patterns of human-AI interaction profiles in team problem-solving interactions to inform the development of guidelines for the effective use of AI in team problem-solving interactions.

To sum up, human-machine collaboration in complex problem-solving is a rapidly growing area of research that has the potential to significantly improve the efficiency and effectiveness of problem-solving in a wide range of practical applications. Some key areas of focus for research in this area include.

1. Developing more advanced machine learning and artificial intelligence techniques to enable machines to better understand and interpret human inputs, and to more effectively collaborate with humans in problem-solving tasks.
2. Investigating new forms of human-machine interaction, such as natural language processing and gesture-based interfaces, to make it easier for humans to interpret, communicate with and control machines.
3. Developing new methods for evaluating the performance of human-machine teams, in order to better understand the strengths and weaknesses of different forms of collaboration and to identify areas for improvement.
4. Exploring the ethical and social implications of human-machine collaboration, in order to ensure that these systems are developed and deployed in ways that are fair and beneficial to all stakeholders.
5. Applying human-machine collaboration to real-world complex problems in fields such as healthcare, finance, transportation, and environmental management, as well as education.

Overall, human-machine collaboration in complex problem-solving has the potential to rejuvenise how researchers and practicioners approach and solve some of the most pressing challenges facing society today.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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